NOT MEASUREMENT SENSITIVE
MIL-L-18331B(AS)
20 February 1990
SUPERSEDING
MIL-L-18331A(AS)
27 April 1978

MILITARY SPECIFICATION

LEAD ALLOY PIG

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers pigs of one type of lead alloy used in the making of forming dies.

2. APPLICABLE DOCUMENTS

This section is not applicable to this specification.

3. REQUIREMENTS

3.1 Chemical composition. The chemical composition of the lead alloy pigs, when determined in accordance with 4.6, shall conform with Table I.

TABLE I. Chemical composition

Element	Analysis (%)
Antimony	5.0-7.0
Copper (max)	0.25
Other impurities (max total)	0.25
Lead	Remainder

3.2 Weight. Each pig shall weigh from 25 to 100 pounds.

3.3 Workmanship. The pigs shall be clean and of uniform composition, free from segregations, dross, oxides, foreign material, or other injurious defects.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Systems Engineering and Standardization Department (Code 53), Naval Air Engineering Center, Lakehurst, NJ 08733-5100 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A
FSC 9650
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:26Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The examination and testing of the lead alloy pigs shall be quality conformance (acceptance) inspection as specified in this section.

4.3 Examination of product. Each alloy pig shall be examined visually to determine apparent conformance with this specification with respect to the requirements for workmanship (3.3).

* 4.4 Sampling for chemical analysis. One pig shall be selected from every ton or fraction thereof in a lot. For purposes of sampling, a lot shall consist of all pigs from the same cast from a single, homogeneous heat or melt of metal. Any addition of raw material to the heat or melt shall constitute a new lot.

4.5 Preparation of chemical analysis samples. If more than one lot (4.4) is included in a shipment, each lot shall be sampled and analyzed separately. For each lot, the sample pigs selected for tests shall be broken so as to present a clean fracture over the cross section of the pigs. One-half of each pig shall have drillings taken from three points triangularly spaced on the clean part of the fracture. One of the three drilling points shall be located half way from the center of the cross section to the bottom of the pig; another point shall be located half way between the center of the cross section and a top corner of the pig. The other point shall be located half way between the center of the cross section and other top corner of the pig. An equal proportion of drillings shall be taken from each hole in each sample pig of the lot. These drillings shall be combined, thoroughly mixed, and an analysis shall be made from the mixed drillings. All drillings shall be fine, clean, dry, and free from foreign substances. Drillings shall be taken at slow speed and shall not be oxidized.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:26Z
Check the source to verify that this is the current version before use.

4.5.1 Chemical analysis. At least six (6) ounces of the mixed drillings shall be taken for chemical analysis. In case the first analysis shows that the material does not conform to the specification, a check analysis shall be made of the same mixed sample of drillings. The average of these analyses shall be final.

* 4.6 Test methods. Chemical composition may be determined by wet chemical methods (4.6.1) or by spectrochemical methods (4.6.2).

4.6.1 Classical wet chemical methods.

4.6.1.1 Antimony. The percentage of antimony shall be determined as follows: Dissolve 1 gram of drillings in 20 ml conc. H_2SO_4 by heating in a 200 ml Erlenmeyer flask, adding also 2.5 grams of acid potassium sulphate. Keep near boiling, with a cover on the flask, until all the sample is snow-white or nearly so; finally bring to slight boiling of the concentrated acid and continue this boiling for ten minutes until all is snow-white. Add 100 ml water and 10 ml conc. HCl and boil gently for ten minutes to dissolve all antimony away from the lead sulphate. Cool, rinse into a 400 ml beaker and dilute to 300 ml. Cool to 10°C or lower and titrate immediately with 0.1N potassium permanganate solution. Standardize the potassium permanganate solution by putting the U.S. National Bureau of Standards lead-base standard sample No. 53 through these operations.

4.6.1.2 Copper. The percentage copper shall be determined spectrographically as follows: Dissolve an apporopriate weight in a covered 300 ml Erlenmeyer flask in 25 ml tartaric acid solution (500 grams dissolved in 1000 ml distilled water) and 5 ml conc. HNO_3. When action has ceased, add 20 ml tartaric acid solution and 4 ml conc. HNO_3. Continue digesting, and when solution is complete, cool, transfer to a graduated cylinder, dilute to appropriate volume with tartaric acid solution and mix thoroughly. Transfer duplicate 0.1 ml portions to recessed electrodes, dry at 100°C to 150°C for 30 minutes, and estimate the copper spectrographically.

4.6.1.3 Other impurities. Excessive amounts of other elements present as impurities may be noted in the spectrogram.

* 4.6.2 Spectrochemical methods. Percentages of antimony, copper, and other impurities may be determined using spectrochemical methods in accordance with method 112.2 of FED-STD-151, or as otherwise approved by the contracting activity.

4.6.3 Lead. The percentage of lead shall be taken by difference.

4.7 Rejection criteria. Individual lead alloy pigs which fail to meet the requirements for workmanship (3.3) in the examination of product (4.3) shall be rejected. If the results of the first chemical analysis (4.5.1) of the mixed sample drillings from any lot fail to conform with the requirements of Table I and the averaged results of the first analysis and the check analysis (4.5.1) also fail to conform with those requirements, the lot shall be rejected.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:26Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packing. Unless otherwise specified, lead alloy pigs shall be packed and palletized in such a manner as to permit handling by mechanical equipment and insure acceptance by common or other carrier for safe transportation at the lowest rate to the point of delivery.

5.2 Marking. Each pig shall be stenciled with the following information:

Lead Antimony Alloy
Specification MIL-L-18331
Weight
Name of Contractor
Order Number

6. NOTES

* (This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

* 6.1 Intended use. The forming dies produced from the lead alloy pigs will be used in the manufacture of sheet metal for stampings.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification
b. Quantity (weight) of lead alloy required

* 6.3 Subject term (keyword) listing.

Lead
Lead Alloy
Lead Alloy Pig
Pig

* 6.4 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Preparing activity:
Navy - AS
(Project No. 9650-N041)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:26Z
Check the source to verify that this is the current version before use.

INSTRUCTIONS: In a continuing effort to make our standardization documents better, the DoD provides this form for use in submitting comments and suggestions for improvements. All users of military standardization documents are invited to provide suggestions. This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and mailed. In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was too rigid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the problems. Enter in block 6 any remarks not related to a specific paragraph of the document. If block 7 is filled out, an acknowledgement will be mailed to you within 30 days to let you know that your comments were received and are being considered.

NOTE: This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document*(s)* or to amend contractual requirements.

(Fold along this line)

(Fold along this line)

DEPARTMENT OF THE NAVY
Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 53
Lakehurst, NJ 08733-5100

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO. 12503 WASHINGTON D.C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE NAVY

Commanding Officer
Naval Air Engineering Center
Systems Engineering and Standardization Department
(SESD), Code 53
Lakehurst, NJ 08733-5100

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:26Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-L-18331B(AS)	LEAD ALLOY PIG

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:26Z
Check the source to verify that this is the current version before use.

(TO DETACH THIS FORM, CUT ALONG THIS LINE.)